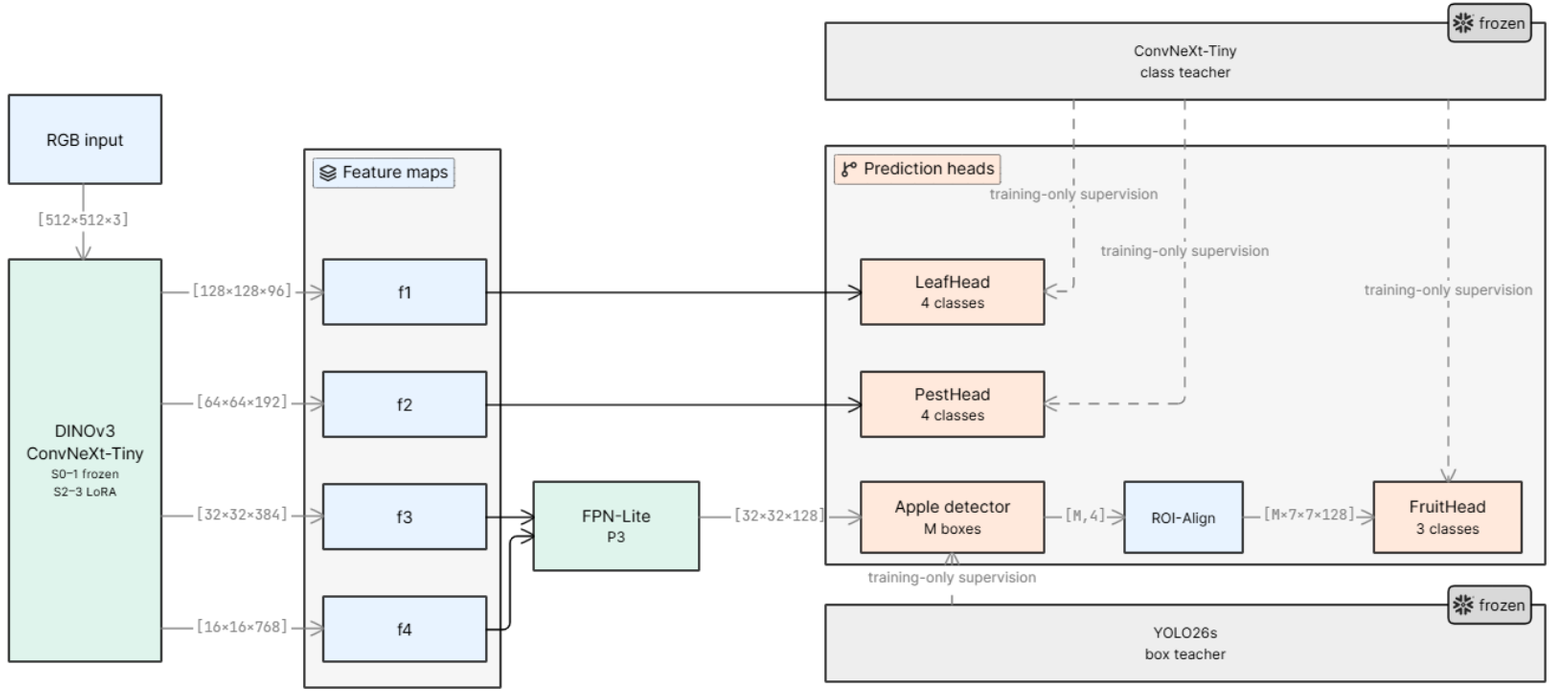
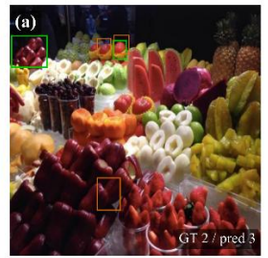
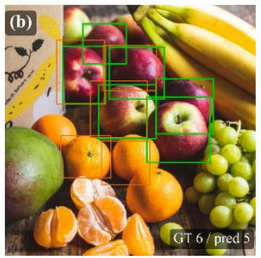
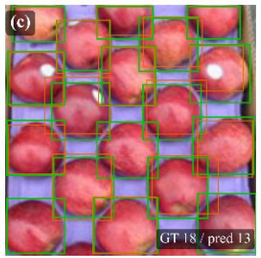
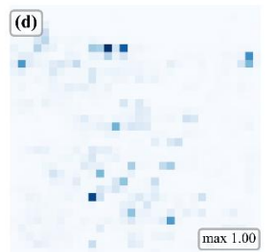
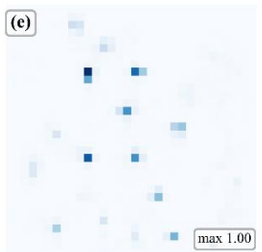

RGB input
[512×512×3]
DINOv3
ConvNeXt-Tiny
S0–1 frozen
S2–3 LoRA
[128×128×96]
[64×64×192]
[32×32×384]
[16×16×768]
Feature maps
f1
f2
f3
f4
FPN-Lite
P3
[32×32×128]
frozen
ConvNeXt-Tiny
class teacher
Prediction heads
training-only supervision
training-only supervision
training-only supervision
LeafHead
4 classes
PestHead
4 classes
Apple detector
M boxes
[M,4]
ROI-Align
[M×7×7×128]
FruitHead
3 classes
training-only supervision
frozen
YOLO26s
box teacher

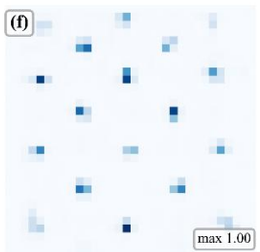

(a)
GT 2 / pred 3


(b)
GT 6 / pred 5


(c)
GT 18 / pred 13


(d)
max 1.00


(e)
max 1.00


(f)
max 1.00